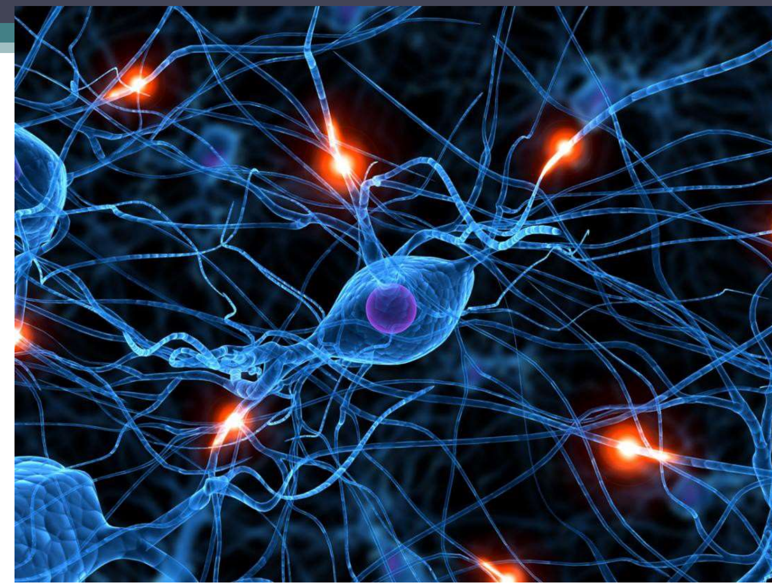
Unit 1 –
Introduction to Neural Networks

Human Brain



- Complex, Non-linear,
  Parallel Computer
- Capability to organize its structural constituents – Nervous (Neurons)
- Example :
  - Human Vision – Information processing task – perceptual recognition tasks that takes 100 to 200ms

Image source : Medium

Human Vision (Perceptual Recognition)

- Identify the Objects in the image
  - Car
  - Telephone
  - Trolly
  - Mouse
  - Radio
  - Hammer
  - Doll
  - Clips
  - Etc.,

Image Source : Dreamstime

Human Brain (contd.,)

- Considerable structure, ability to build up its own rule of behavior through experience
  - $1 + 2 + 3 = ??$
  - $1 + 2 + 3 + 4 + \ldots., + 10 = ??$
  - $1^2 + 2^2 + 3^2 == ??$
  - $1^2 + 2^2 + 3^2 + \ldots\ldots + 10^2 = ??$

Neural (Nervous) Network

- Developing a neuron (nervous) system is a synonym of a plastic brain – *Plasticity*
- Developing nervous system to adapt to its surrounding environment
- *Neural Network –* machine designed in a way in which the brain performs a task of interest.

Neural Network (NN)

- Massively parallel distributed processor made up of simple processing units that has a natural propensity for storing experiential knowledge and making it available for use.
  - Knowledge is acquired by the network from its environment through a learning process
  - Interneuron connection strengths known as synaptic weights are used to store the acquired knowledge.

Learning algorithm

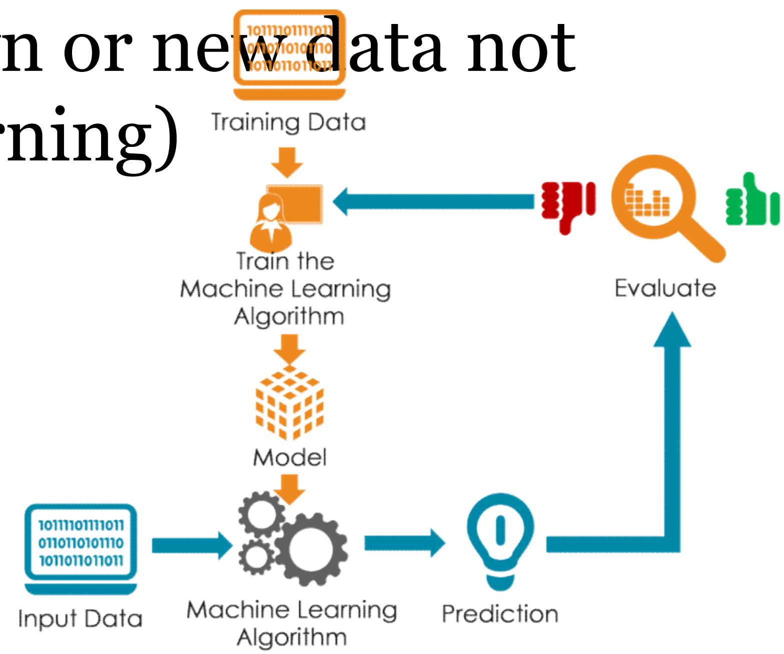
- Procedure used to perform the learning process
- Function of which is to modify the synaptic weights of the network in an orderly fashion to attain a desired design objective.

2 x 2 = 4							
2 x 3 = 6	3 x 3 = 9						
2 x 4 = 8	3 x 4 = 12	4 x 4 = 16					
2 x 5 = 10	3 x 5 = 15	4 x 5 = 20	5 x 5 = 25				
2 x 6 = 12	3 x 6 = 18	4 x 6 = 24	5 x 6 = 30	6 x 6 = 36			
2 x 7 = 14	3 x 7 = 21	4 x 7 = 28	5 x 7 = 35	6 x 7 = 42	7 x 7 = 49		
2 x 8 = 16	3 x 8 = 24	4 x 8 = 32	5 x 8 = 40	6 x 8 = 48	7 x 8 = 56	8 x 8 = 64	
2 x 9 = 18	3 x 9 = 27	4 x 9 = 36	5 x 9 = 45	6 x 9 = 54	7 x 9 = 63	8 x 9 = 72	9 x 9 = 81

Source : Vedic Maths

Benefits of Neural Networks

- Massively parallel distributed structure
- Ability to learn and generalize
- Generalization – Unknown or new data not available in training (Learning)

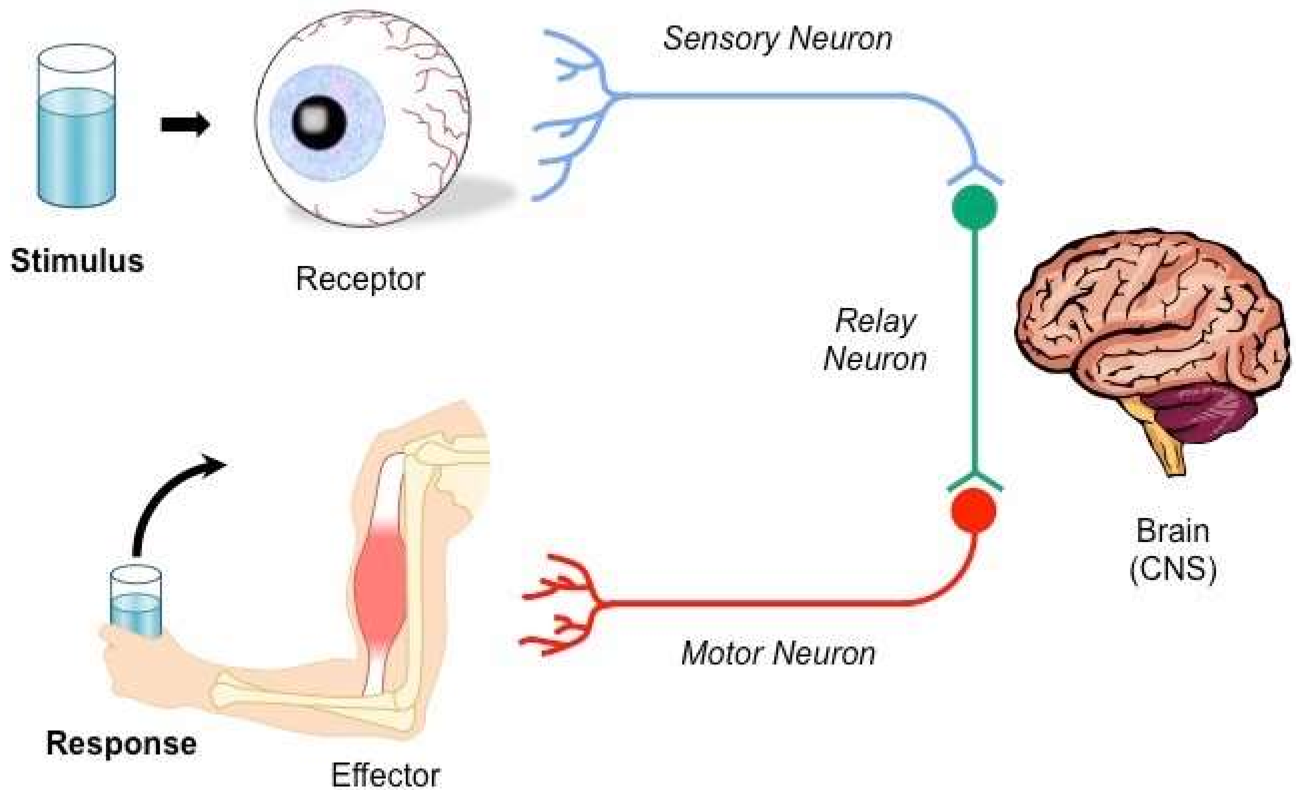


Source : Medium

Benefits of Neural Networks

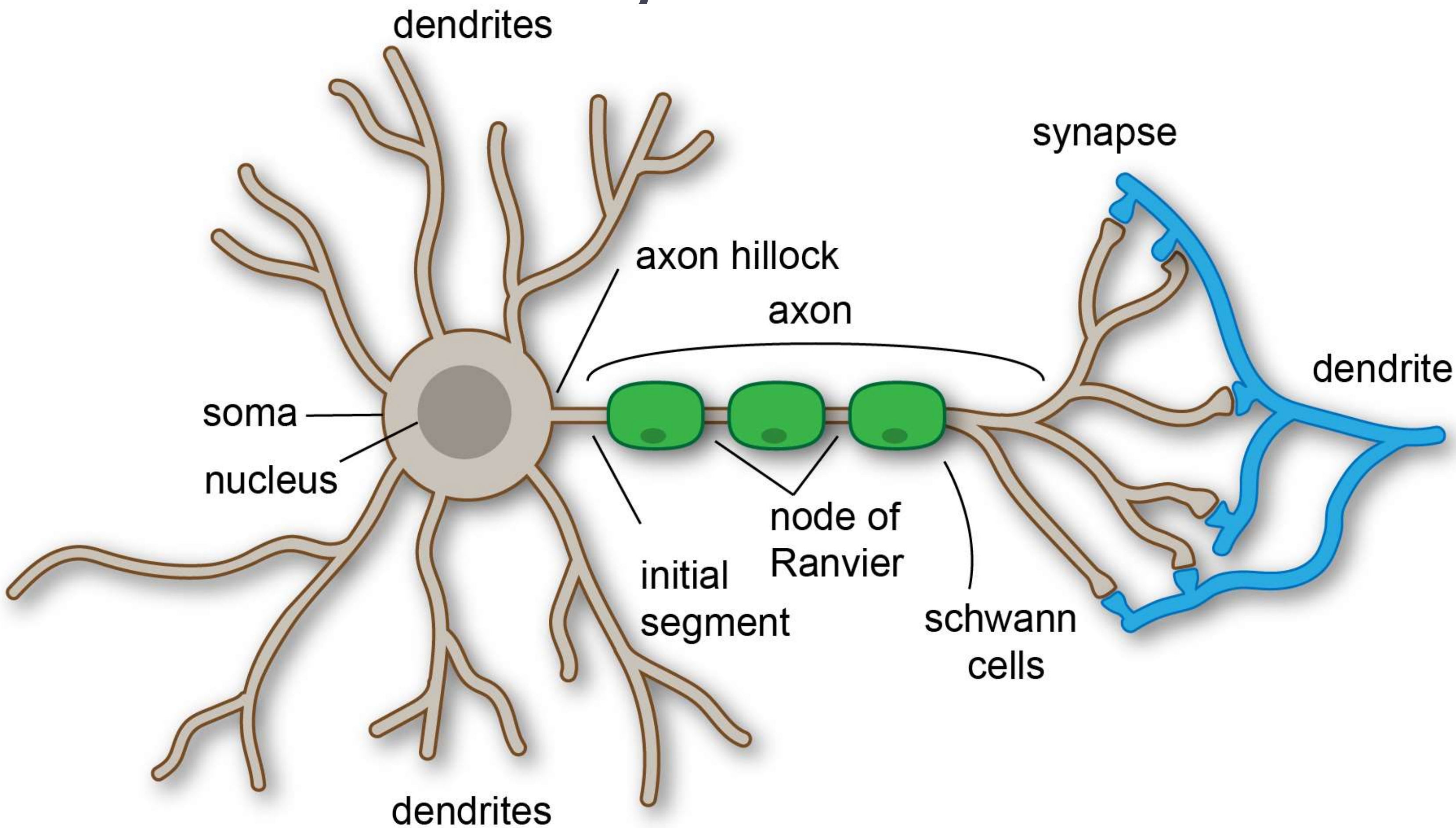
- Non-linearity
- Input–Output Mapping
- Adaptivity
- Evidential response
- Contextual Information
- Fault Tolerance
- Neurobiological Analogy

Human Brain system



Source : Bioindace

Human Brain System



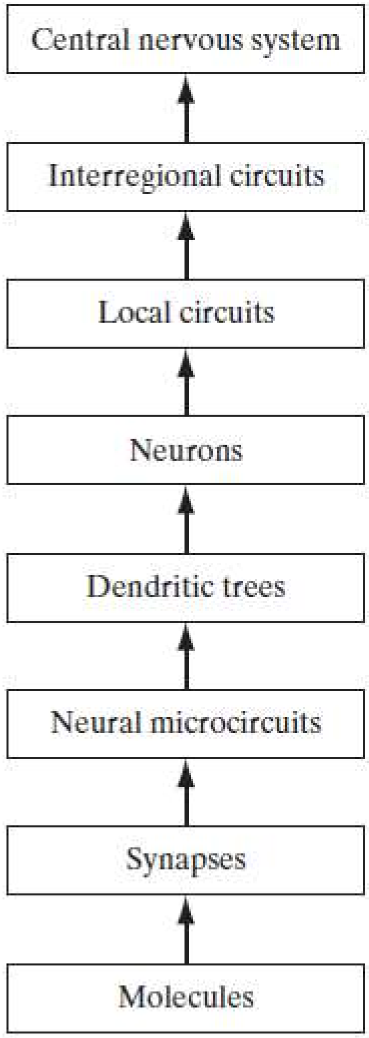
Source: Wikipedia

Human Brain System

- Axons – Transmission lines [smoother surface, fewer branches, greater length]
- Dendrites – Receptive Zones [Irregular surface and more branches] – used to receive inputs from the environments

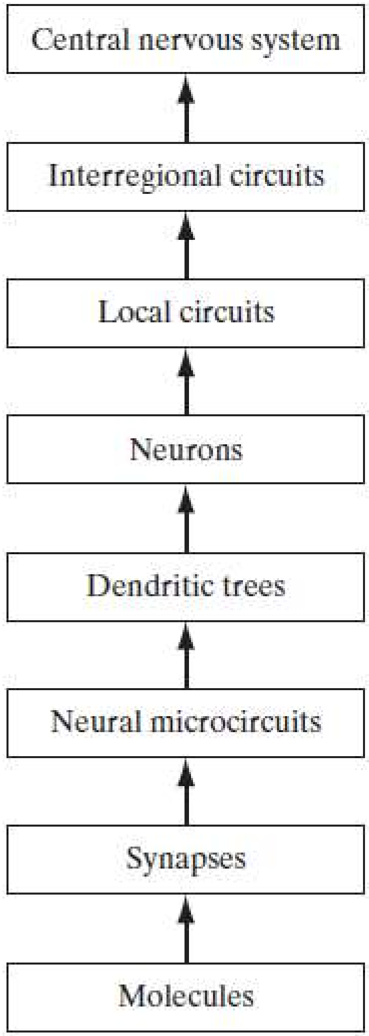
Human Brain system

- synapses - the most fundamental level, depending on molecules and ions for their action
- neural microcircuit - assembly of synapses organized into patterns of connectivity to produce a functional operation of interest. Likened to a silicon chip made up of assembly of transistors [micro-circuits measured in micometeres].
- neural microcircuits are grouped to form dendritic subunits within the dendritic trees of individual neurons
- local circuits (about 1 mm in size) made up of neurons with similar or different properties;



Human Brain system

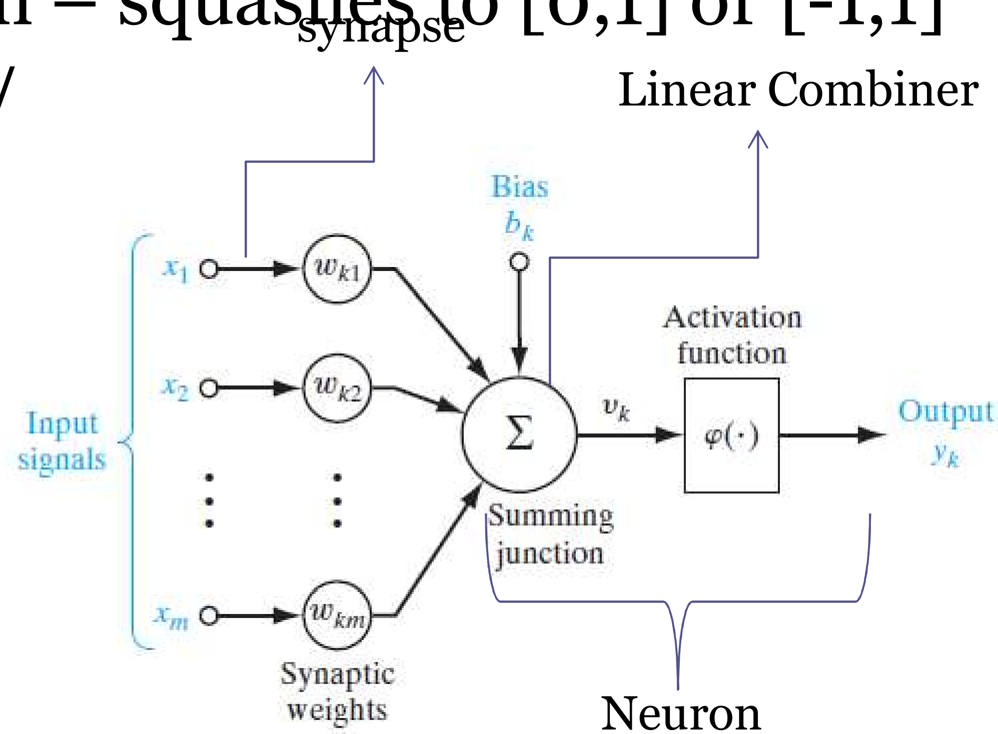
- interregional circuits made up of pathways, columns, and topographic maps, which involve multiple regions located in different parts of the brain.
- Topographic maps are organized to respond to incoming sensory information. These maps are often arranged in sheets, as in the superior colliculus
- Topographic maps and other interregional circuits mediate specific types of behavior in the central nervous system.



A Neuron Model

- Neuron – Information processing unit – fundamental to the operation of a neural network
- Activation function – squashes to [0,1] or [-1,1]
- Bias – To increase/ decrease the net input of the activation function

$$u_k = \sum_{j=1}^{m} w_{kj} x_j$$



A Neuron Model

$$y_k = \varphi(u_k + b_k)$$

$$y_k = \varphi(v_k)$$

$$u_k = \sum_{j=1}^{m} w_{kj} x_j$$

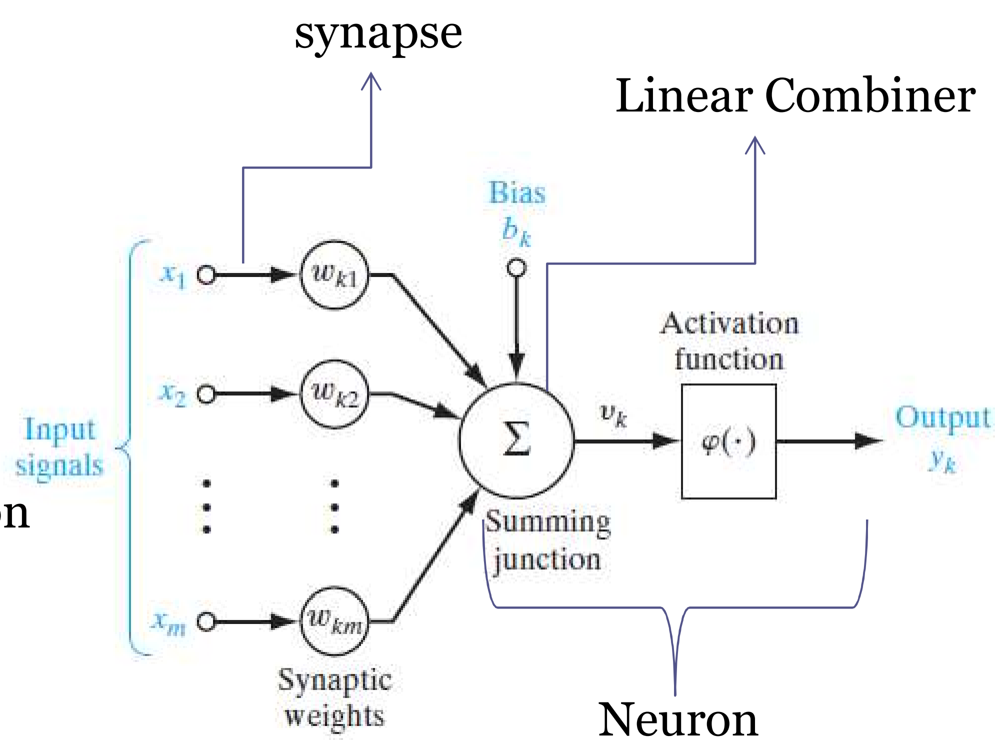
Inputs $x_1, x_2, \ldots, x_m$

Synaptic Weights

$$w_{k1}, w_{k2}, \ldots, w_{km}$$

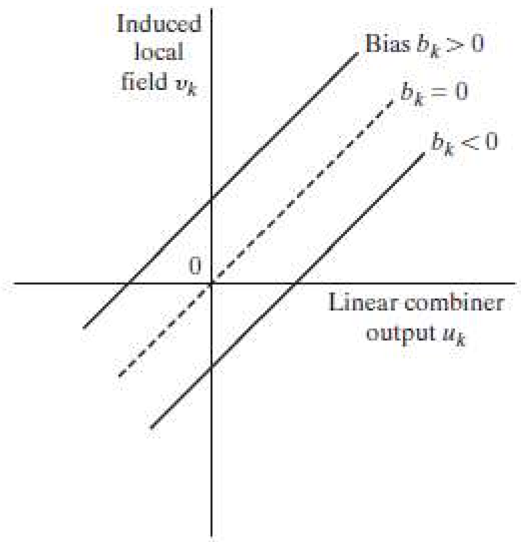
Bias b_k - affine transformation

$$v_k = u_k + b_k$$



A Neuron Model

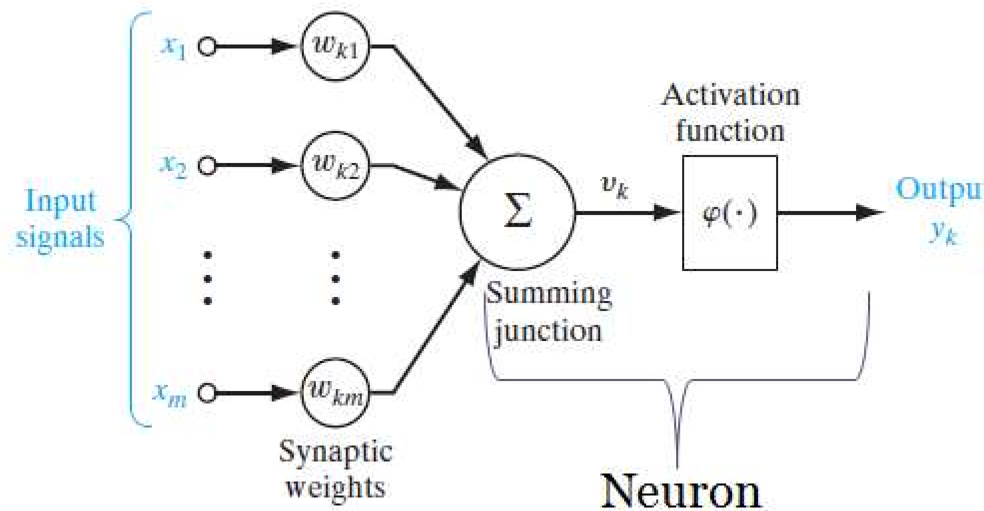
- Affine transformation produced by the presence of a bias



A Neuron Model

$$v_k = \sum_{j=1}^{m} w_{kj} x_j$$

$$y_k = \varphi(v_k)$$

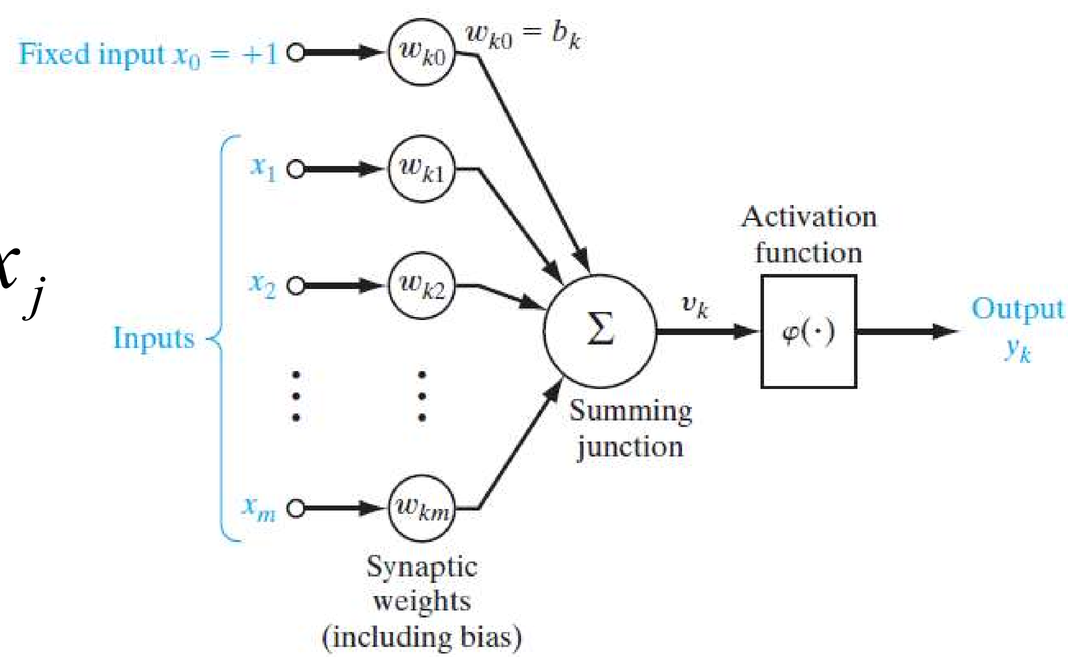


A Neuron Model

- Consider $x_0 = 1$ and $w_{k0} = b_k$

$$v_k = 2b_k + \sum_{j=1}^{m} w_{kj} x_j$$

$$y_k = \varphi(v_k)$$



Activation function

- Defines the output of a neuron in terms of induced local field v_k
  - Threshold function
  - Piecewise linear function
  - Sigmoid function

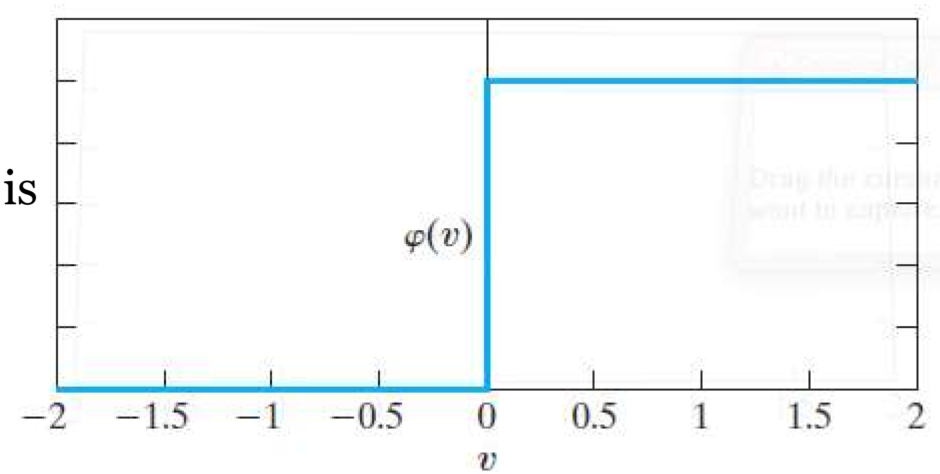
Threshold (Heaviside) function

The Local field on activation function is

$$\varphi(v) = \begin{cases} 1 & v \geq 0 \\ 0 & v < 0 \end{cases}$$

The Output on activation function is

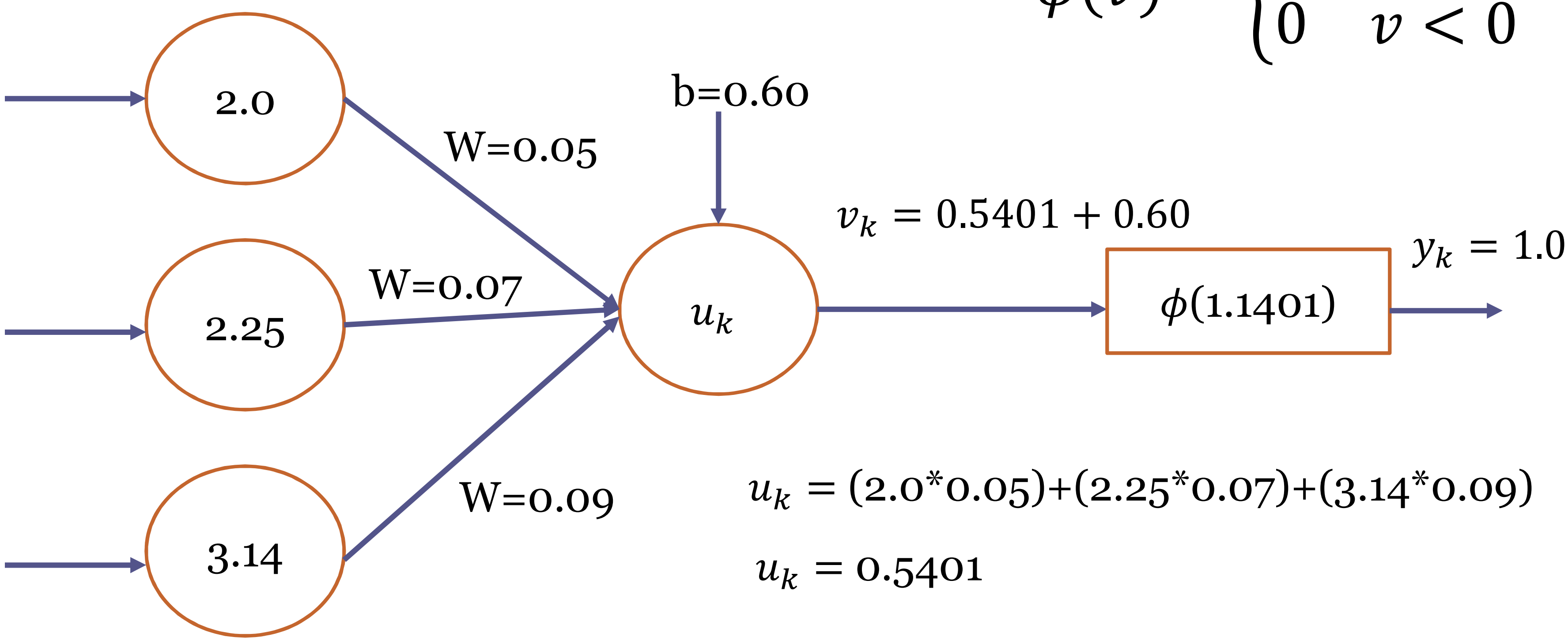
$$y_k = \begin{cases} 1 & v_k \geq 0 \\ 0 & v_k < 0 \end{cases}$$



Neuron is referred to as McCulloch-Pitts Model – Output of a neuron takes on the value of 1 if the induced local field is non negative and 0 otherwise

Threshold (Heaviside) function

$$\phi(v) = \begin{cases} 1 & v \geq 0 \\ 0 & v < 0 \end{cases}$$



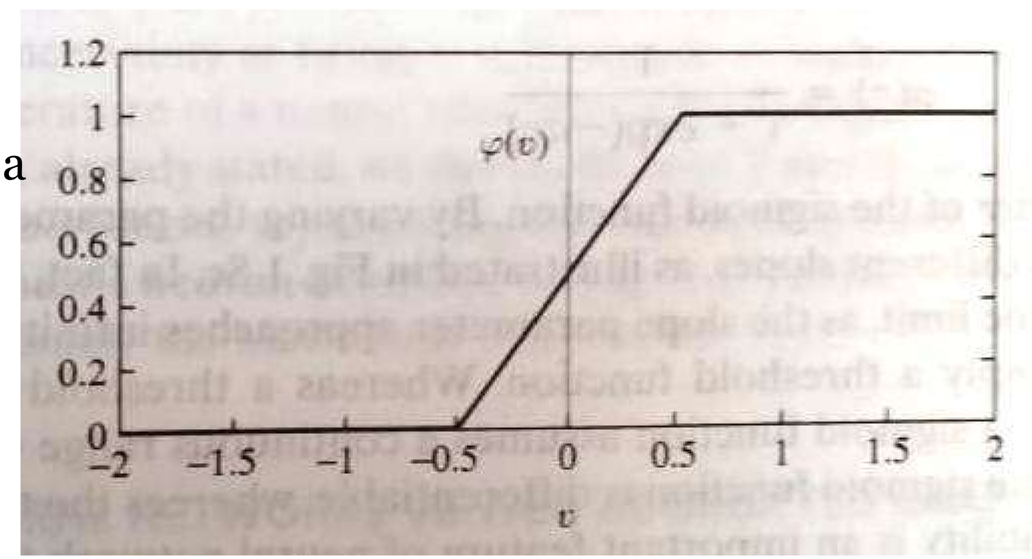
$$u_k = (2.0*0.05)+(2.25*0.07)+(3.14*0.09)$$

$$u_k = 0.5401$$

Piecewise linear function

$$\varphi(v) = \begin{cases} 1 & v \geq +\frac{1}{2} \\ v & +\frac{1}{2} > -\frac{1}{2} \end{cases} \qquad \varphi(v) = 0 \quad v \leq -\frac{1}{2}$$

Assumed to be **approximation** to a
Non-linear amplifier

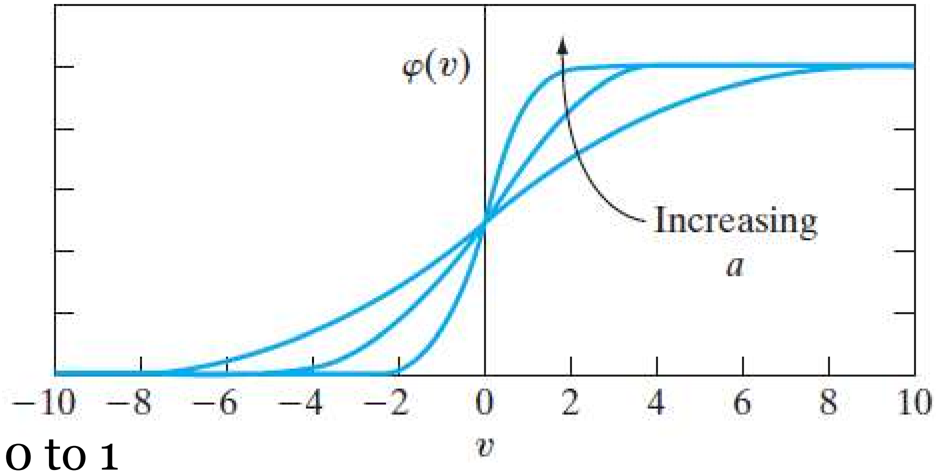


Sigmoid Function

- S –shaped graph – common activation function used in the neural network model .
- Strictly increasing function exhibits a graceful balance between linear and non-linear behavior.

$$\phi(v) = \frac{1}{1 + e^{-av}}$$

a – slope parameter



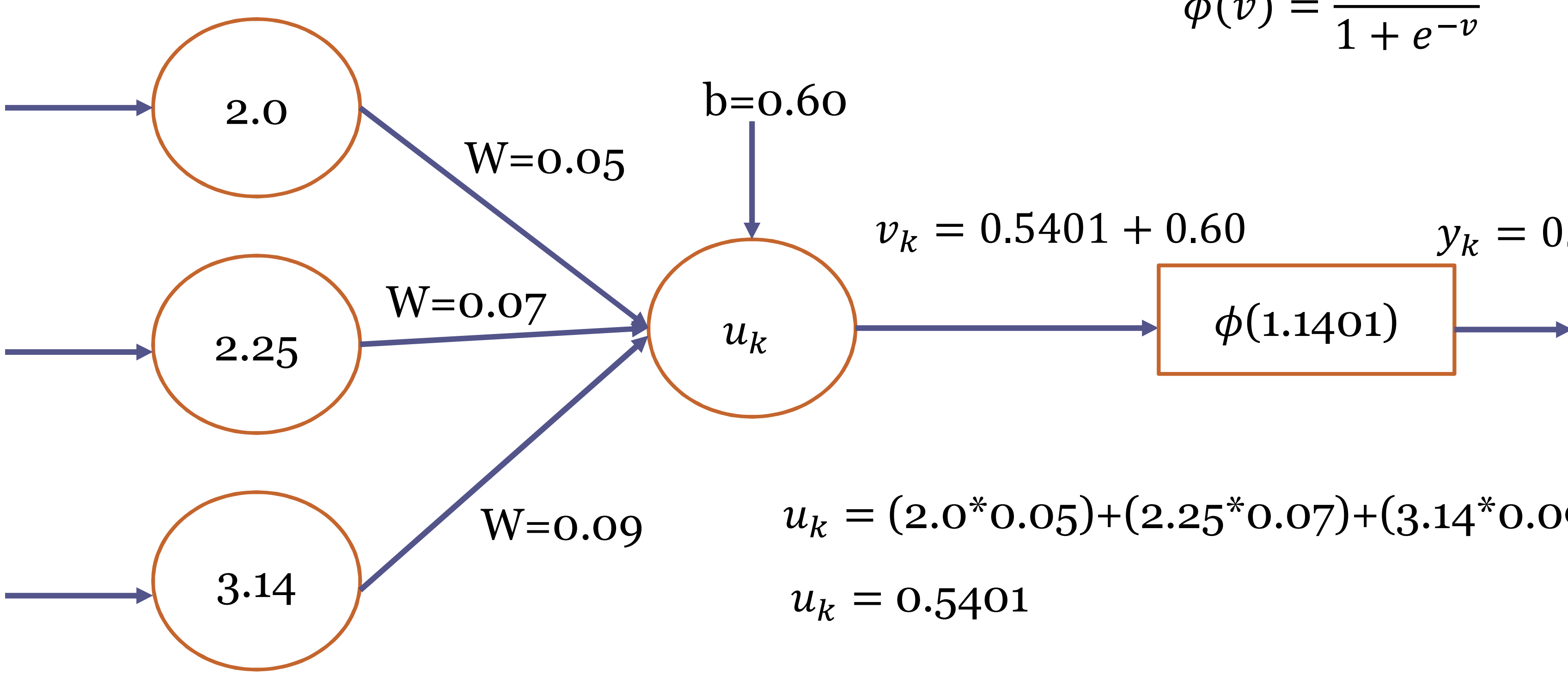
Note : Threshold function -> 0 or 1
Sigmoid function -> continuous value of 0 to 1

Sigmoid function

- At $v = 0$ $\qquad \phi(v) = \frac{1}{1+e^{0}} = \frac{1}{1+1} = 0.5$
- At $v = 1$ $\qquad \phi(v) = \frac{1}{1+e^{-1}} = \frac{1}{1+0.36} = 0.73$
- At $v = 5$ $\qquad \phi(v) = \frac{1}{1+e^{-5}} = \frac{1}{1+0.0067} = 0.993$
- At $v = -1$ $\qquad \phi(v) = \frac{1}{1+e^{-(-1)}} = \frac{1}{1+3.718} = 0.211$
- At $v = -5$ $\qquad \phi(v) = \frac{1}{1+e^{-(-5)}} = \frac{1}{1+148} = 0.006$

Sigmoid function

$$\phi(v) = \frac{1}{1 + e^{-v}}$$



$$u_k = (2.0*0.05)+(2.25*0.07)+(3.14*0.09)$$

$$u_k = 0.5401$$

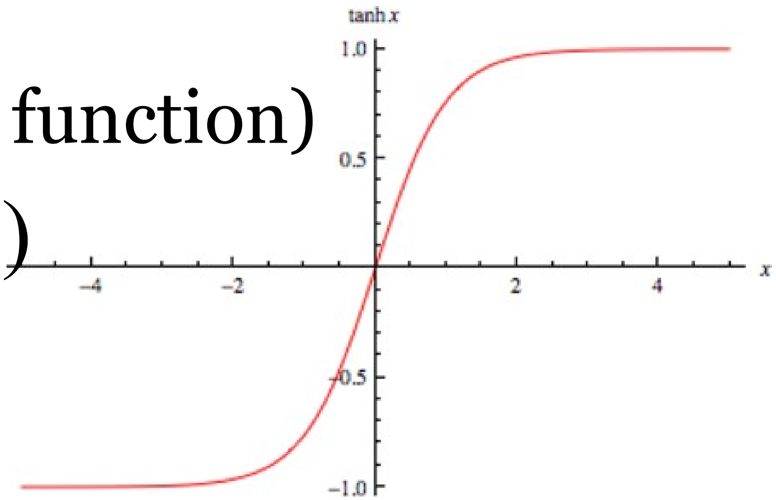
Sigmoid function

- To have the activation function from range -1 to +1
- Threshold function

$$\varphi(v) = \begin{cases} 1 & v > 0 \\ 0 & v = 0 \\ -1 & v < 0 \end{cases}$$

- Sigmoid (hyperbolic tangent function)

$$\varphi(v) = \tanh(v)$$



Stochastic model (Neuron)

- McCulloch–Pitts model with states resides only +1 or -1
- Consider an Air conditioner - State of a Switch (ON or OFF) – To Switch on or not

$$x = \begin{cases} +1 & P(v) \\ -1 & 1 - P(v) \end{cases}$$

$$P(v) = \frac{1}{1+e^{-\frac{v}{T}}}$$

T is the pseudo temperature – used to control the AC and therefore uncertainty in firing

THANK YOU